

White Paper

Antibody-Drug Conjugates in Cancer Therapy

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NEHA RANI, SENIOR INSIGHTS ASSOC, ARK PATENT INTELLIGENCE, IQVIA
TARISHI GUPTA, SENIOR INSIGHTS ASSOC, ARK PATENT INTELLIGENCE, IQVIA

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Introduction

Cancer remains one of the most formidable health challenges, affecting millions of lives across every corner of the globe.[1] The World Health Organisation's 2024 report recorded 20 million new cancer cases and 9.7 million deaths in 2022, with projections reaching 35 million cases by 2050.[2] Conventional cancer treatment options were quite systemic and non-specific, leading to side effects. Over decades, a remarkable shift towards more targeted and personalized therapies have led to the use of antibody-drug conjugates (ADCs) as a promising therapy for cancer.[1]

ADCs are advanced therapies that link monoclonal antibodies (mAbs) to potent cytotoxic drugs, known as payloads, via cleavable or non-cleavable linkers. They deliver drugs directly to cancer cells by recognizing specific antigens, thereby minimizing harm to healthy tissue, and enabling a bystander effect that kills nearby tumour cells.[1] Despite their promise, ADCs are complex and costly to develop, with challenges, including off-target toxicity and drug resistance. Currently, 15 ADCs are approved for various cancers, such as breast, blood, gastric, urothelial, cervical, and lung cancers (Figure 1),[1] and around 200 are in clinical trials, including 24 in Phase III, all driven by advances in antibody, linker, and payload technologies.[3]

Today, ADCs have proven to be game changers in cancer therapy with growing market needs.[1] The first ADC drug approval, Pfizer's MylotargTM (Gemtuzumab ozogamicin) by the US Food and Drug Administration (FDA) in 2000 for treating acute myeloid leukaemia (AML), marked a milestone in precision oncology.[4]

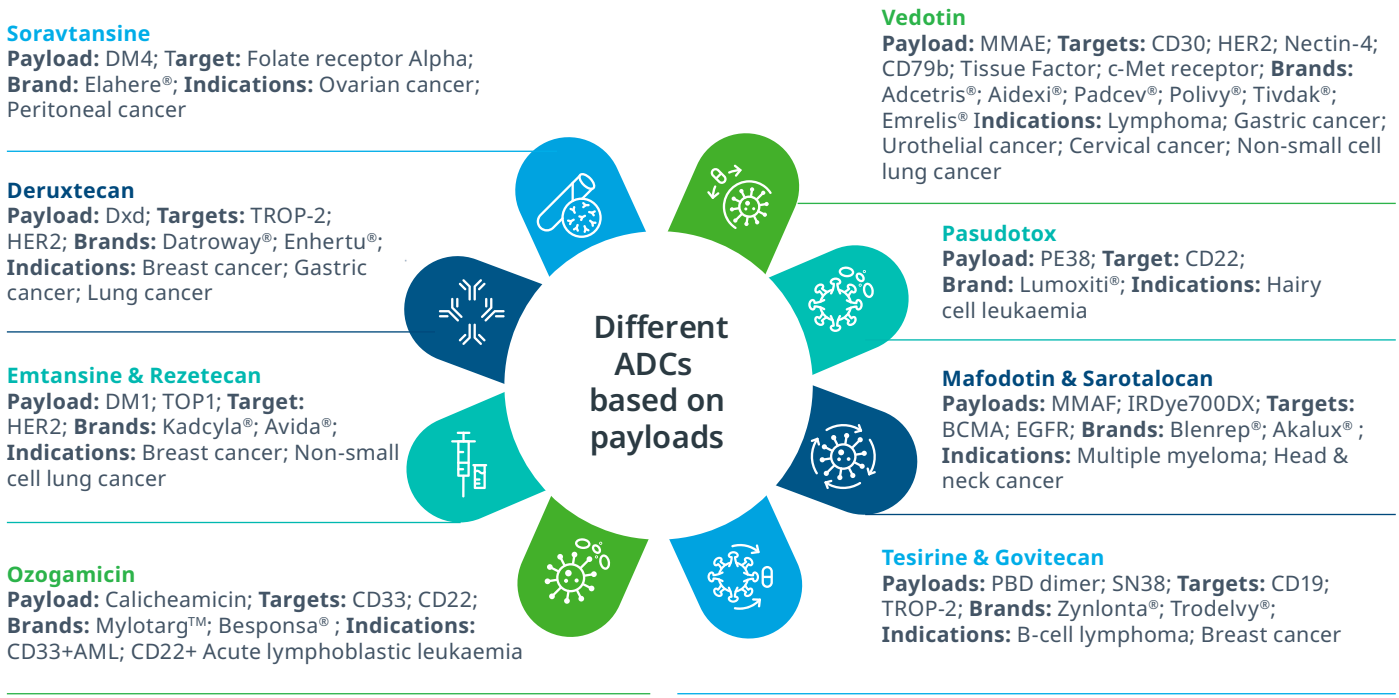
The European Medicines Agency (EMA) followed with approval in 2018, granting it orphan drug status due to the rarity of AML.[5] The USA ADC market, valued at USD 11.3 billion in 2023, is projected to grow rapidly in the upcoming years.[6]

Ark Patent Intelligence offers comprehensive insights into the global ADC drug market. It provides extensive worldwide information on drug approvals, market insights, key patents, regulatory protections, litigation trends, and detailed in-depth analyses of marketed, registered and phase III drugs covering more than 30 ADCs.

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USA ADC MARKET IN 2023 — AND GROWING RAPIDLY

Despite their promise, ADCs are complex and costly to develop, with challenges, including off-target toxicity and drug resistance. Currently, 15 ADCs are approved for various cancers, such as breast, blood, gastric, urothelial, cervical, and lung cancers and around 200 are in clinical trials.

Figure 1: Different types of marketed ADCs with payloads, targets, brands, and indications



Notes: Different payloads used - MMAE: Monomethyl auristatin E; PE38: A 38kD fragment of Pseudomonas exotoxin A; MMAF: Monomethyl auristatin-F; IRDye700DX: Near-infrared phthalocyanine dye; PBD dimer: Pyrrolobenzodiazepine; SN38: Active metabolite of Irinotecan; DM1: Mertansine or Emtansine, derivative of maytansine 1; TOP1- Topoisomerase I; DXd: Exatecan derivative for ADC; DM4: Ravtansine, a modified form of maytansine.

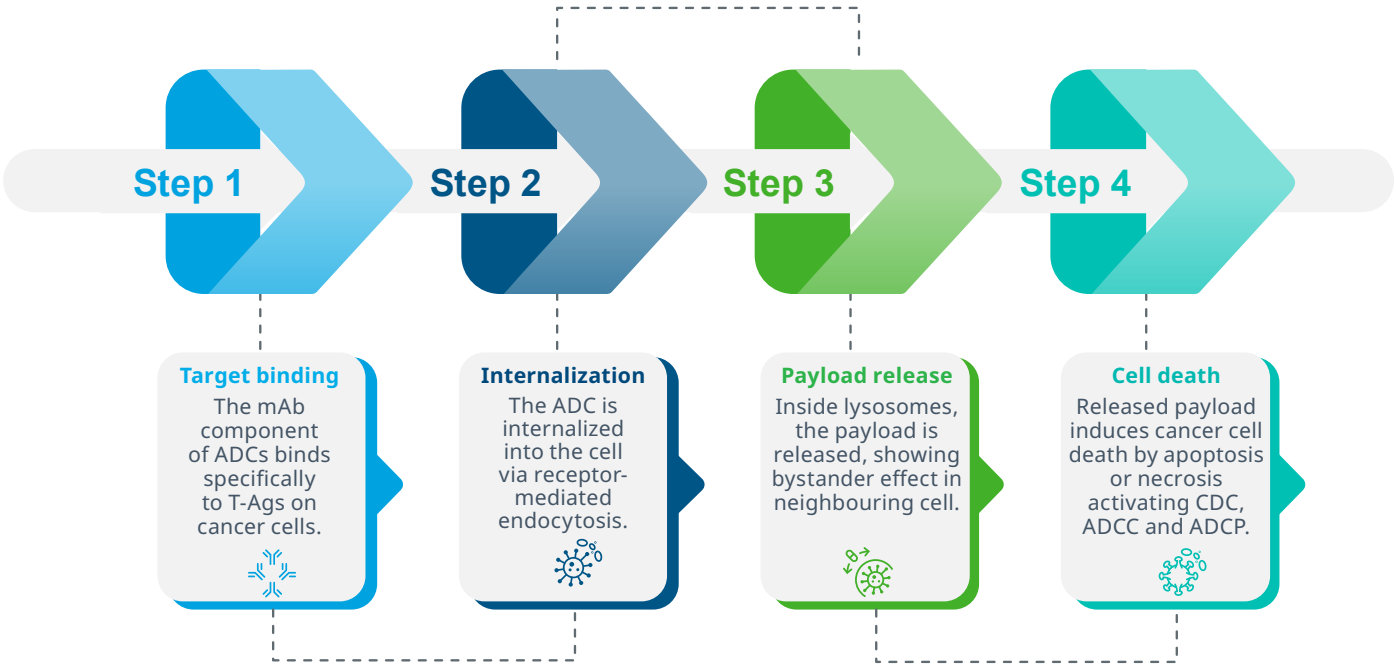
Different targets used – CD: Cluster of differentiation of various cell surface molecules - CD30, CD79b, CD22, CD19 and CD33; c-Met receptor- Mesenchymal-Epithelial Transition Factor; HER2- Human Epidermal growth factor Receptor 2; BCMA- B Cell Maturation Antigen; EGFR- Epidermal Growth Factor Receptor; TROP-2- Trophoblast cell surface antigen 2.

Structural insights on ADCs and their mechanism of action

ADCs consist of three key components a mAb, a linker, and a payload.[1,3] The mAb component identifies Tumour-Associated antigens (T-Ags) via its Fragment antigen binding (Fab) region and can trigger immune responses such as Complement-Dependent Cytotoxicity (CDC), Antibody-Dependent Cellular Cytotoxicity (ADCC), and Antibody-Dependent Cellular Phagocytosis (ADCP), inducing cell death through apoptosis or necrosis (Figure 2).[7,8] Linkers in ADCs connect the antibody to the payload, enabling controlled release.

Over 70% use cleavable linkers that respond to acidic or enzymatic environments in lysosomes, like those in Mylotarg and Enhertu®. Non-cleavable linkers, seen in Kadcyla®, offer greater plasma stability and release the payload only after cancer cell internalization.[1,3,4] The cytotoxic payload is the core active component of ADCs, spanning various drug classes. Tubulin binders are the most widely used, present in 53.3% of approved ADCs and 45.8% of Phase III candidates. Other payloads include DNA-damaging agents and Topoisomerase I (TOP1) inhibitors. A key design metric in ADC is the Drug-to-Antibody Ratio (DAR), typically ranging between 0 and 8 in approved ADCs, balancing potency with safety.[3]

Figure 2: A four-step mechanism of action of ADCs



ADC development

ADCs have evolved through four generations, each improving specificity, stability, and potency as described in (Table 1).[1] With advancement in hybridoma technology in 1975, Paul Ehrlich used murine antibodies and basic linkers in early ADCs, laying the foundation for newer ADCs. Second- and third-generation ADCs introduced humanized antibodies, site-specific conjugation, and potent payloads.

The latest, fourth-generation ADCs, like Trodelvy® (Sacituzumab govitecan), feature high DARs, novel payloads, and dual-targeting strategies, enhancing efficacy even in resistant tumours. Despite the progress, challenges such as complex conjugation processes and resistance to payloads still remain.[1,3]

Table 1: Development of ADCs

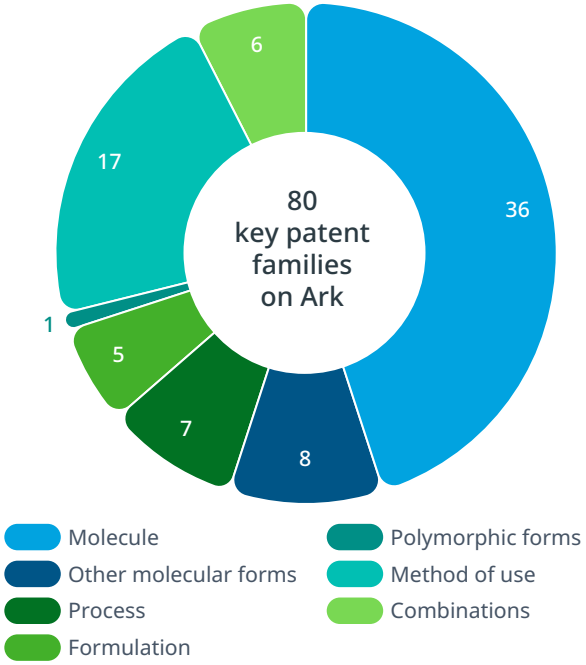
ADC GENERATION	TIMELINE	ANTIBODIES	DAR	ADVANTAGE	DISADVANTAGE	EXAMPLES
1st	Before 2000	Murine or chimeric	0 to 8	Proof of concept, basic targeting	Poor stability, off-target toxicity	Mylotarg; esponsa
2nd	2000–2015	Humanized	4 to 8	Improved linkers, better safety	Limited efficacy, resistance issues	Kadcyla; Adcetris
3rd	2015–2020	Fully humanized	2 to 4	Site-specific conjugation, potent payloads	Complex manufacturing, high cost	Enhertu
4th	2020 – present	Multi-specific	7 to 8	Smart payloads, controlled release, multi-targeting	Technical challenges, limited clinical data	Trodelvy

ADC landscape via Ark Patent Intelligence

According to IQVIA Ark Patent Intelligence, the global patent landscape for ADCs has expanded significantly in recent years. As of August 2025, there are 17 marketed and 1 registered ADCs worldwide, supported by around 80 key patent families and with over 36 families alone for molecule patents (Figure 3).

Ark Patent Intelligence provides extensive worldwide information on drug approvals, market insights, key patents, regulatory protections, litigation trends, and detailed in-depth analyses of marketed, registered and phase III drugs.

Figure 3: Different ADC patient types vs number of families on Ark Patient Intelligence





Case study 1

Strategic insights into Kadcyla and Enhertu

Kadcyla and Enhertu are both HER2-targeting ADCs derived from Trastuzumab, but they differ in their payloads and linkers. Genentech's Kadcyla (Trastuzumab emtansine) was approved in 2013 in both the USA and Europe and uses the DM1 payload with a stable linker for the treatment of breast cancer[9,10] In contrast, Enhertu (Trastuzumab deruxtecan) by Daiichi Sankyo, approved in 2019 in the USA and in 2021 in Europe uses, the DXd payload with a cleavable linker, and treats multiple cancers including breast, lung, gastric, and solid tumours.[7,11]

According to Ark Patent Intelligence, as of August 2025, Kadcyla had exclusivity protections in the USA until February 2025 and in Europe until November 2023, with respective key patents expiring between 2015 and 2028. This means that there is nothing blocking biosimilar Trastuzumab emtansine from being approved; however, it cannot be marketed until 2026 in the USA and 2028 in Europe, at the earliest.

In contrast, biosimilar Trastuzumab deruxtecan will not be approved until 2029 in Europe and 2031 in the USA, at the earliest, and Enhertu enjoys longer patent protection, with patents valid until December 2033 and January 2036 in the USA and Europe, respectively (Table 2). This highlights a promising opportunity for pharmaceutical companies to innovate new ADCs through varied combinations of antibodies, linkers, and payloads. For example, while the original Trastuzumab is no longer under patent protection, companies are progressing with innovative ADC variants in their pipeline, such as Trastuzumab botidotin, Trastuzumab duocarmazine, Trastuzumab brengitecan, etc. Compared to manufacturing biosimilars of existing ADCs, developing a new ADC using a well-established antibody tends to be more financially advantageous. Additionally, these innovative products can secure strong patent coverage and extended legal protections, making it harder for biosimilar competitors to enter the market.

Table 2: Selected patent and exclusivities expiries in the USA and Europe for Kadcyla and Enhertu

ACTIVE INGREDIENT	REGION	BRAND	APPROVAL DATE	EXCLUSIVITY TYPE	EXCLUSIVITY EXPIRY	PATENT TYPE	PATENT EXPIRY	LOSS OF PROTECTION
Trastuzumab emtansine	USA	Kadcyla	Feb 2013	BASE	Feb 2017	Use patent-US7097840	Jul 2026	Jul 2026
	Europe		Nov 2013	NCE	Nov 2021	Molecule patent + SPC EP1689846	Nov 2028	Nov 2028
Trastuzumab deruxtecan	USA	Enhertu	Dec 2019	BASE	Dec 2023	Molecule patent with PTE US10155821	Dec 2033	Dec 2033
	Europe		Jan 2021	NCE	Jan 2029	Molecule patent + SPC EP3101032	Jan 2036	Jan 2036

Source: Ark Patent Intelligence

Notes: BASE: Biosimilar Application Submission Exclusivity in the USA; NCE: New chemical entity data exclusivity in Europe; SPC: Supplementary protection certificate in Europe; PTE: Patent Term Extension in the USA.

Case study 2

Strategic insights into Adcetris and Polivy

Adcetris® (Brentuximab vedotin) and Polivy® (Polatuzumab vedotin) are leading ADCs in haematologic oncology. Both use the payload MMAE, which disrupts microtubules and induces cell death. Adcetris, developed by Seagen, was approved in 2011 and 2012 in the USA and Europe, respectively, for Hodgkin's Lymphoma (HL) and systemic anaplastic large cell lymphoma.[12,13] Polivy, by Genentech, was approved in 2019 in the USA and in 2020 in Europe for diffuse large B-cell lymphoma (DLBCL).[14,15]

According to Ark Patent Intelligence, Adcetris was protected by exclusivity in Europe until October 2023 and in the USA until February 2024, with key patents expiring between 2026 and 2027 (Table 3). On the other hand, Polivy will maintain its exclusivity in Europe until January 2028 and in the USA until June 2031. Its USA molecule patent will lose its protection in August 2029, while the European molecule patent, extended by SPC, will remain valid until July 2033.

This shows that there is an opportunity for biosimilar Brentuximab vedotin to be approved, even though it cannot be marketed until 2026 in the USA and 2027 in Europe, at the earliest. In contrast, biosimilar Polatuzumab vedotin will not be approved until 2028 in Europe and 2031 in the USA, at the earliest, and Polivy enjoys longer patent protection, with USA patents valid until June 2031 and European patents until July 2033.

The upcoming expiration of the above mentioned patents is expected to pave the way for biosimilar competition; however, developing a new ADC by combining different mAbs with the same linker and payload can lead to innovative treatments such as Telisotuzumab vedotin and Enfortumab vedotin. This approach could significantly shift market dynamics for innovator companies, who would then enjoy novel product protections.

Table 3: Different patent and exclusivities expiries in the USA and Europe for Adcetris and Polivy

ACTIVE INGREDIENT	REGION	BRAND	APPROVAL DATE	EXCLUSIVITY TYPE	EXCLUSIVITY EXPIRY	PATENT TYPE	PATENT EXPIRY	LOSS OF PROTECTION
Brentuximab vedotin	USA	Adcetris	Aug 2011	BASE	Feb 2016	Molecule patent-US7659241	Jul 2026	Jul 2026
	Europe		Oct 2012	NCE	Oct 2020	Molecule patent + SPC EP1545613	Oct 2027	Oct 2027
Polatuzumab vedotin	USA	Polivy	Jun 2019	BASE	Jun 2023	Molecule patent US8722857	Aug 2029	Jun 2031
				BAAE	Jun 2031			
	Europe		Jan 2020	NCE	Jan 2028	Molecule patent + SPC EP2474557	Jul 2033	Jul 2033

Source: Ark Patent Intelligence

Notes: PED: Paediatric exclusivity; BAAE: Biosimilar Application Approval Exclusivity.

Conclusions

ADCs are advancing cancer treatment by offering targeted therapies with proven success in breast, gastric, and blood cancers. Blockbuster drugs like Enhertu and Kadcyla highlight their commercial viability. Despite challenges such as off-target toxicity, drug resistance, and high manufacturing costs, advancements in molecular design and stability continue to improve their efficacy. According to Ark Patent Intelligence, Kadcyla faces earlier patent expiry by 2026 to 2028 in the USA and Europe, unlike Enhertu, which retains protection until 2033 to 2036 in both regions. In the treatment of lymphomas, Adcetris and Polivy present viable therapeutic options. But Adcetris could face biosimilar competition post-2027, while Polivy enjoys longer protection. The expanding patent landscape and diverse antibody-payload combinations present new opportunities for pharmaceutical companies and would bring renewed hope to patients.

The expanding patent landscape and diverse antibody-payload combinations present new opportunities for pharmaceutical companies and would bring renewed hope to patients.

References

1. https://www.nature.com/articles/s41392-022-00947-7#Sec8
2. https://www.who.int/news/item/01-02-2024-global-cancer-burden-growing--amidst-mounting-need-for-services
3. https://jhoonline.biomedcentral.com/articles/10.1186/s13045-025-01704-3
4. https://www.accessdata.fda.gov/drugsatfda_docs/label/2020/761060s004lbl.pdf
5. https://www.ema.europa.eu/en/medicines/human/EPAR/mylotarg
6. https://pubmed.ncbi.nlm.nih.gov/39037635/
7. https://www.accessdata.fda.gov/drugsatfda_docs/label/2025/761139s032s035lbl.pdf
8. https://www.iqvia.com/library/white-papers/breaking-new-ground-advancing-cancer-care-with-novel-therapeutic-modalities
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14. https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/761121s008lbl.pdf
15. https://www.ema.europa.eu/en/documents/product-information/polivy-epar-product-information_en.pdf

About the authors



NEHA RANI
Senior Insights Associate,
IQVIA

Neha Rani holds a master's degree in pharmaceutical chemistry from Kurukshetra University, Haryana, India. She is a Senior Insight Associate at IQVIA and has over 7 years of experience in patent searches and patent analytics. She has been a part of patent intelligence team since 2023.



TARISHI GUPTA
Senior Insights Associate,
IQVIA

Tarishi Gupta holds a master's degree in biotechnology from Indian Institute of Technology, Guwahati, India. She is a Senior Insights Associate at IQVIA and has nearly 6 years of experience in patent searches and analysis. She has been a part of patent intelligence team since 2019.

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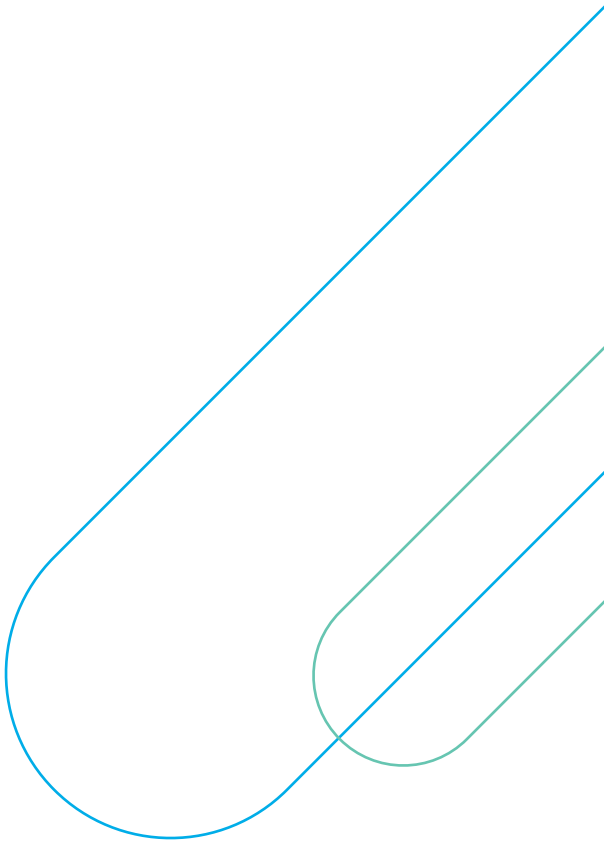
CONTACT US

PatentIntelligence@iqvia.com

X: @IQVIA_GMI

LinkedIn: IQVIA Global Market Insights

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